



MEMORY MAP

GCEM40 Series In-Situ Stack Gas Analysers

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-	Issue : A	Rev. : 1	Date : 7/1/11	Doc. i/d : 0116/6	Ref. : 090046
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CODEL International Ltd is a UK company based in the heart of the Peak District National Park at Bakewell, Derbyshire. The company specialises in the design and manufacture of high-technology instrumentation for the monitoring of combustion processes and atmospheric pollutant emissions.

The constant search for new products and existing product improvement keeps CODEL one step ahead. With a simple strategy, to design well-engineered, rugged, reliable equipment, capable of continuous operation over long periods with minimal maintenance, CODEL has set standards both for itself and for the rest of the industry.

All development and design work is carried out 'in-house' by experienced engineers using proven state-of-the-art CAD and software development techniques, while stringent assembly and test procedures ensure that the highest standards of product quality, synonymous with the CODEL name, are maintained.

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GCEM40 Series Memory Map

CONTROLLER MASTER

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7E00

Memory Map: Internal EEROM :- Cross sensitivity parameters

Location	Description	
7E00 }	a	02 }
7E01 }		44 }
7E02 }	b	00 }
7E03 }		AA }
7E04 }	a	05 }
7E05 }		71 }
7E06 }	b	00 }
7E07 }		92 }
7E08 }	a	02 }
7E09 }		E2 }
7E0A }	b	07 }
7E0B }		68 }
7E0C }	c	0E }
7E0D }		09 }
7E0E }	d	1C }
7E0F }		BC }
7E10 }	a	06 }
7E11 }		4B }
7E12 }	b	0C }
7E13 }		42 }
7E14 }	c	0F }
7E15 }		E9 }
7E16 }	d	2B }
7E17 }		CA }
7E18 }	a	03 }
7E19 }		06 }
7E1A }	b	08 }
7E1B }		02 }
7E1C }	c	0F }
7E1D }		57 }
7E1E }	d	1E }
7E1F }		ED }

CO2 on NO2

a=580

b=170

CO2 on SO2

a=1393

b=146

a=738

b=1896

CH4 on N2O

c=-3593

d=7356

a=1611

b=3138

CH4 on NO2

c=-4073

d=11210

a=774

b=2050

CH4 on SO2

c=-3927

d=7917

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7E20

Memory Map: Internal EEROM :- Cross sensitivity parameters

Location	Description			
7E20 }	a	00 }		a=141
7E21 }		8D }		
7E22 }	b	02 }		b=587
7E23 }		4B }		
7E24 }	c	01 }	H2O on CH4	c=328
7E25 }		48 }		
7E26 }	d	00 }		d=91
7E27 }		5B }		
7E28 }	a	02 }		a=736
7E29 }		E0 }		
7E2A }	b	13 }		b=5102
7E2B }		EE }		
7E2C }	c	0F }	H2O on NO2	c=-3967
7E2D }		7F }		
7E2E }	d	07 }		d=1843
7E2F }		33 }		
7E30 }	a	00 }		a=102
7E31 }		66 }	H2O on SO2	
7E32 }	b	00 }		b=175
7E33 }		AF }		
7E34 }	a	03 }	NO2 on CH4	a=913
7E35 }		91 }		
7E36 }	a	04 }	NO2 on N2O	a=1141
7E37 }		75 }		
7E38 }	a	04 }	NO2 on SO2	a=1124
7E39 }		64 }		
7E3A				
7E3B				
7E3C }		00	H2O on NO span 000dec	
7E3D }		00		
7E3E }		03	H2O on CO2 span 900dec	
7E3F }		84		

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7E60

Memory Map: Internal EEROM :- Scale Shapes

Location	Description		
7E60	00	}	CO2 Y target (FFFFh = purged with N2)
7E61	64	}	
7E62	27	}	CO2 A
7E63	10	}	
7E64	00	}	CO2 B
7E65	00	}	
7E66			
7E67	00		Temperature deviation from 130 C for H2O on NO (01 = +1C, FF = -1C)
7E68	0D	}	a = 3361]
7E69	21	}	]]
7E6A	EE	}	b = 4550]
7E6B	3A	}	]]
7E6C	F4	}	c = 2959]
7E6D	71	}	]]
7E6E	08	}	d = 2135]
7E6F	57	}	]]
7E70			auto zero? Not implemented
7E71			
7E72			
7E73			
7E74			
7E75			
7E76			
7E77			
7E78			
7E79			
7E7A			
7E7B			
7E7C			
7E7D			
7E7E			
7E7F			

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7E80

Memory Map: Internal EEROM :- D Cals

Location	Description	
7E80 }		}
7E81 }	D1	}
7E82 }		}
7E83 }		}
7E84 }	D2	}
7E85 }		}
7E86 }		}
7E87 }	D3	}
7E88 }		}
7E89 }		}
7E8A }	D4	}
7E8B }		}
7E8C }		}
7E8D }	D5	}
7E8E }		}
7E8F }		}
7E90 }	D6	}
7E91 }		}
7E92 }		}
7E93 }	D7	}
7E94 }		}
7E95		
7E96		
7E97		
7E98		
7E99		
7E9A		
7E9B		
7E9C		
7E9D		
7E9E		
7E9F		

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7EA0

Memory Map: Internal EEROM :- Set Cals

Location	Description	
7EA0 }		}
7EA1 }	Ch1	}
7EA2 }		}
7EA3 }		}
7EA4 }	Ch2	}
7EA5 }		}
7EA6 }		}
7EA7 }	Ch3	}
7EA8 }		}
7EA9 }		}
7EAA }	Ch4	}
7EAB }		}
7EAC }		}
7EAD }	Ch5	}
7EAE }		}
7EAF }		}
7EB0 }	Ch6	}
7EB1 }		}
7EB2		
7EB3		
7EB4		
7EB5		
7EB6 }	Heater plate	}
7EB7 }		}
7EB8 }	Lens holder	}
7EB9 }		}
7EBA }	PT100	}
7EBB }		}
7EBC }	Flue (Ext'l)	}
7EBD }		}
7EBE		
7EBF		

Set Cals

Cal temperatures

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7EC0

Memory Map: Internal EEROM Normalising Inputs

Location	Description		
7EC0 03 }	Input scaling factor	]	Temperature input parameters
7EC1 E8 }		]	
7EC2 00 }	Lower limit	]	
7EC3 00 }		]	
7EC4 01 }	Upper limit	]	
7EC5 2C }		]	0-10, 0-20, 2-10, 2-20, 4-20mA
7EC6 04 }	Type	]	
7EC7 0A }	Smoothing constant	]	
7EC8 03 }	Input scaling factor	]	Oxygen input parameters
7EC9 E8 }		]	
7ECA 00 }	Lower limit	]	
7ECB 00 }		]	
7ECC 00 }	Upper limit	]	
7ECD FA }		]	0-10, 0-20, 2-10, 2-20, 4-20mA
7ECE 04 }	Type	]	
7ECF 0A }	Smoothing constant	]	
7ED0 03 }	Input scaling factor	]	Pressure input parameters
7ED1 E8 }		]	
7ED2 00 }	Lower limit	]	
7ED3 00 }		]	
7ED4 00 }	Upper limit	]	
7ED5 A0 }		]	0-10, 0-20, 2-10, 2-20, 4-20mA
7ED6 04 }	Type	]	
7ED7 0A }	Smoothing constant	]	
7ED8			
7ED9			
7EDA			
7EDB			
7EDC			
7EDD			
7EDE			
7EDF 0A	Smoothing constant PT100 temperature input		

Model: GCEM42x0 Controller Software ID: 507-055B Module: MM7EE0

Memory Map: Internal EEROM

Location	Description
7EE0 1B }	AWC Temperature target value
7EE1 58 }	
7EE2 40	Proportional constant
7EE3 08	Integral constant
7EE4 01	Control flag (01 for auto, 00 for manual)
7EE5 0A	Time constant n
7EE6 80 }	M/S value
7EE7 00 }	
7EE8 11 }	AWC temperature lower limit (degC * 100)
7EE9 94 }	
7EEA 21 }	AWC temperature upper limit (degC * 100)
7EEB 34 }	
7EEC 01 }	AWC control temperature source
7EED 68 }	
7EEE	
7EEF	
7EF0 12	Ch.1 detector locations (hi nibble D1 active, low nibble D2 ref)
7EF1 35	Ch.2 detector locations
7EF2 25	Ch.3 detector locations
7EF3 45	Ch.4 detector locations
7EF4 FF	Ch.5 detector locations
7EF5 FF	Ch.6 detector locations
7EF6 FF	
7EF7 FF	
7EF8 03	Ch.1 definition
7EF9 06	Ch.2 definition
7EFA 0A	Ch.3 definition
7EFB 02	Ch.4 definition
7EFC FF	Ch.5 definition
7EFD FF	Ch.6 definition
7EFE FF	
7EFF FF	

Channel Definitions

01	=	CO
02	=	CO2
03	=	NO
06	=	SO2
0A	=	H2O

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7F00

Memory Map: Internal EEROM

Location		Description	
7F00			
7F01			
7F02			
7F03			
7F04			
7F05			
7F06			
7F07			
7F08			
7F09			
7F0A	32	Ch1	Z smoothing factors
7F0B	32	Ch2	
7F0C	32	Ch3	
7F0D	32	Ch4	
7F0E	32	Ch5	
7F0F	32	Ch6	
7F10			
7F11			
7F12			
7F13			
7F14			
7F15			
7F16			
7F17	00	-ve ppm or % values: 00=zero, 01=allow	
7F18	64	Ch1	ppm scaling factors
7F19	64	Ch2	
7F1A	64	Ch3	
7F1B	64	Ch4	
7F1C	64	Ch5	
7F1D	64	Ch6	
7F1E			
7F1F		Reference EEROM Location	

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7F20

Memory Map: Internal EEROM

Location	Description		
7F20			
7F21			
7F22			
7F23			
7F24			
7F25			
7F26			
7F27			
7F28			
7F29			
7F2A			
7F2B			
7F2C			
7F2D			
7F2E			
7F2F			
7F30	07	}	Path Length mm
7F31	D0	}	
7F32			
7F33			
7F34			
7F35			
7F36			
7F37			
7F38	1E		Weather cover tolerance (1E = 3.0degC)
7F39	0A		Pressure input tolerance (0A = 10kPa)
7F3A			(reserved for Det Heatsink temperature tolerance)
7F3B			
7F3C			
7F3D			
7F3E	3C		Power on period before GCEM cal (1unit = 1minute)
7F3F	1E		Plant on period before GCEM cal (1unit = 1minute)

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7F40

Memory Map: Internal EEROM :-

Location			Description	
7F40	03	}	Ch1	}
7F41	E8	}		}
7F42	03	}	Ch2	}
7F43	E8	}		}
7F44	03	}	Ch3	}
7F45	E8	}		}
7F46	03	}	Ch4	}
7F47	E8	}		}
7F48	03	}	Ch5	}
7F49	E8	}		}
7F4A	03	}	Ch6	}
7F4B	E8	}		}
7F4C				
7F4D				
7F4E				
7F4F				
7F50	02	}	Purge counter value	}
7F51	58	}		}
7F52	0C		Purge flag (01=power up,02=plant off,	] Parameters
7F53	00		0C=precal,08=in cal)	] used
7F54	01		Calibration initiation flag 01= activate	] for
7F55	64		No. Cal cycles (cal counter will be 8*)	] timed
7F56	3C		After Cal delay (delay will be 10*secs.)	] calibrations
7F57				]
7F58	01	}	Auto span : pre purge time	}
7F59	2C	}		}
7F5A	02	}	Auto span : span gas time	] Parameters
7F5B	58	}		] used for
7F5C	01	}	Auto span : post purge time	] auto span
7F5D	2C	}		]
7F5E				
7F5F				

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7F60

Memory Map: Internal EEROM :- ATC Values

Location	Description		
7F60 }	Ch1	}	ATC Heater Plate Temp
7F61 }		}	
7F62 }	Ch2	}	
7F63 }		}	
7F64 }	Ch3	}	
7F65 }		}	
7F66 }	Ch4	}	
7F67 }		}	
7F68 }	Ch5	}	
7F69 }		}	
7F6A }	Ch6	}	
7F6B }		}	
7F6C			
7F6D			
7F6E			
7F6F			
7F70 }	Ch1	}	ATC Lens Holder Temp
7F71 }		}	
7F72 }	Ch2	}	
7F73 }		}	
7F74 }	Ch3	}	
7F75 }		}	
7F76 }	Ch4	}	
7F77 }		}	
7F78 }	Ch5	}	
7F79 }		}	
7F7A }	Ch6	}	
7F7B }		}	
7F7C			
7F7D			
7F7E			
7F7F			

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7F80

Memory Map: Internal EEROM :-

Location	Description		
7F80 }	Ch1 }	ATC PT100 Temp	
7F81 }			
7F82 }	Ch2 }		
7F83 }			
7F84 }	Ch3 }		
7F85 }			
7F86 }	Ch4 }		
7F87 }			
7F88 }	Ch5 }		
7F89 }			
7F8A }	Ch6 }		
7F8B }			
7F8C			
7F8D			
7F8E			
7F8F			
7F90 }	Ch1 }	ATC External flue gas Temp	
7F91 }			
7F92 }	Ch2 }		
7F93 }			
7F94 }	Ch3 }		
7F95 }			
7F96 }	Ch4 }		
7F97 }			
7F98 }	Ch5 }		
7F99 }			
7F9A }	Ch6 }		
7F9B }			
7F9C			
7F9D			
7F9E			
7F9F			

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7FA0

Memory Map: Internal EEROM :- Q1, Q2 scaling factors

Location	Description			
7FA0 }	Sq }	Ch1 }	Q1 (X H2O) scaling factors [NO on H2O channel]	
7FA1 }				
7FA2 }	Sq }	Ch2 }		
7FA3 }				
7FA4 }	Sq }	Ch3 }		
7FA5 }				
7FA6 }	Sq }	Ch4 }		
7FA7 }				
7FA8 }	Sq }	Ch5 }		
7FA9 }				
7FAA }	Sq }	Ch6 }	Q2 (X CO2) scaling factors	
7FAB }				
7FAC				
7FAD				
7FAE				
7FAF				
7FB0 }	Sq }	Ch1 }		
7FB1 }				
7FB2 }	Sq }	Ch2 }		
7FB3 }				
7FB4 }	Sq }	Ch3 }		
7FB5 }				
7FB6 }	Sq }	Ch4 }		
7FB7 }				
7FB8 }	Sq }	Ch5 }		
7FB9 }				
7FBA }	Sq }	Ch6 }		
7FBB }				
7FBC				
7FBD				
7FBE				
7FBF				

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7FC0

Memory Map: Internal EEROM :- Q3, Q4 scaling factors

Location Description

7FC0 }	Sq }	Ch1 }	
7FC1 }			
7FC2 }	Sq }	Ch2 }	Q3 (X) scaling factors
7FC3 }			
7FC4 }	Sq }	Ch3 }	[T(K) on CO channel] - Model 4010 only
7FC5 }			[CO on NO channel]
7FC6 }	Sq }	Ch4 }	[NO2 on SO2 channel]
7FC7 }			[NO2 on CH4 channel]
7FC8 }	Sq }	Ch5 }	[NO2 on N2O channel]
7FC9 }			
7FCA }	Sq }	Ch6 }	
7FCB }			
7FCC			
7FCD			
7FCE			
7FCF			
7FD0 }	Sq }	Ch1 }	
7FD1 }			
7FD2 }	Sq }	Ch2 }	
7FD3 }			
7FD4 }	Sq }	Ch3 }	Q4 (X) scaling factors
7FD5 }			
7FD6 }	Sq }	Ch4 }	[CH4 on SO2 channel]
7FD7 }			[CH4 on NO2 channel]
7FD8 }	Sq }	Ch5 }	[CH4 on N2O channel]
7FD9 }			
7FDA }	Sq }	Ch6 }	
7FDB }			
7FDC			
7FDD			
7FDE			
7FDF			

Model: GCEM40x0 Controller Software ID: 507-055B Module: MM7FE0

Memory Map: Internal EEROM :- Q5, A1 scaling factors

Location	Description			
7FE0 }	Sq	}	Ch1	}
7FE1 }		}		}
7FE2 }	Sq	}	Ch2	}
7FE3 }		}		}
7FE4 }	Sq	}	Ch3	}
7FE5 }		}		}
7FE6 }	Sq	}	Ch4	}
7FE7 }		}		}
7FE8 }	Sq	}	Ch5	}
7FE9 }		}		}
7FEA }	Sq	}	Ch6	}
7FEB }		}		}
7FEC				
7FED				
7FEE				
7FEF				
7FF0 }	Sa	}	H2O on CO channel	}
7FF1 }		}		}
7FF2 }	Sa	}	H2O on NO channel	}
7FF3 }		}		}
7FF4 }	Sa	}	H2O on CH4 channel	}
7FF5 }		}		}
7FF6 }	Sa	}	H2O on HCl channel	}
7FF7 }		}		}
7FF8 }	Sa	}	CO2 on CO channel	}
7FF9 }		}		}
7FFA }	Sa	}	CO2 on NO channel	}
7FFB }		}		}
7FFC }	Sa	}	CO2 on CH4 channel	}
7FFD }		}		}
7FFE }	Sa	}	CO2 on HCl channel	}
7FFF }		}		}

Q5 (X) scaling factors
[N2O on SO2 channel]

A scaling factors

HEAD MASTER

Model: GCEM40x0 IC4

Software ID: 507-057A

Module: MM7EE0

Memory Map: Internal EEROM

Location			Description
7EE0	1C	}	Plate Temperature target value
7EE1	20	}	
7EE2	40		Proportional constant
7EE3	08		Integral constant
7EE4	01		Control flag (01 for auto, 00 for manual)
7EE5	0A		Time constant n
7EE6	80	}	M/S value
7EE7	00	}	
7EE8	FF		
7EE9	FF		
7EEA	FF		
7EEB	FF		
7EEC			
7EED			
7EEE			
7EEF			
7EF0	1C	}	Detector heatsink target temperature
7EF1	20	}	
7EF2	40		Proportional constant
7EF3	10		Integral constant
7EF4	00		Differential constant
7EF5	0A		Time constant n
7EF6	80	}	Mark/space control value
7EF7	00	}	
7EF8	01		Control flag (01 for auto, 00 for manual)
7EF9			
7EFA			
7EFB			
7EFC			
7EFD			
7EFE			
7EFF			

Model: GCEM40x0 IC4

Software ID: 507-057A Module: MM7F00

Memory Map: Internal EEROM

Location	Description
7F00 32	D1 detector smoothing factor
7F01 32	D2 detector smoothing factor
7F02 32	D3 detector smoothing factor
7F03 32	D4 detector smoothing factor
7F04 32	D5 detector smoothing factor
7F05 32	D6 detector smoothing factor
7F06 32	D7 detector smoothing factor
7F07 FF	
7F08 10	D1 low detector thresholds
7F09 10	D2 low detector thresholds
7F0A 10	D3 low detector thresholds
7F0B 10	D4 low detector thresholds
7F0C 10	D5 low detector thresholds
7F0D 10	D6 low detector thresholds
7F0E 10	D7 low detector thresholds
7F0F 01	IR source 01 = On, 00 = Off
7F10 0A	Thermistor smoothing factor
7F11 0A	Cooler current smoothing factor
7F12 0A	Source heatsink temperature smoothing factor
7F13 0A	Motor PCB temperature smoothing factor
7F14 0A	Micro PCB temperature smoothing factor
7F15 0A	Heater plate temperature smoothing factor
7F16 0A	Lens holder temperature smoothing factor
7F17 0A	Detector heatsink temperature smoothing factor
7F18 0A	Source voltage smoothing factor
7F19 0A	Cooler drive voltage smoothing factor
7F1A	
7F1B	
7F1C	
7F1D	
7F1E	
7F1F	

Model: GCEM40X0 IC4

Software ID: 507-057A

Module: MM7F20

Memory Map: Internal EEROM

Location			Description
7F20	30	}	Thermistor target value
7F21	00	}	
7F22	32		Proportional constant
7F23	46		Integral constant
7F24	01		Control flag (01 for auto, 00 for manual)
7F25	02		Time constant n
7F26		}	Peltier DAC value
7F27		}	
7F28			Peltier DAC value high byte limit
7F29			
7F2A			
7F2B			
7F2C			
7F2D			
7F2E			
7F2F			
7F30			
7F31			
7F32			
7F33			
7F34			
7F35			
7F36			
7F37			
7F38	32		Thermistor tolerance for date valid
7F39	C8		Heater plate temperature tolerance for data valid (+/- 2.00 C)
7F3A	6B		Lens plate temperature upper limit degC for data valid (67.00 C)
7F3B	32		Source voltage tolerance for data valid (+/- 0.5V)
7F3C			
7F3D			
7F3E			
7F3F			Reference EEROM location (00 = load default parameters at start-up)

HEAD SLAVE

Model: GCEM40X0 IC2 Software ID: 507-056A Module: MM7E00

Memory Map: Internal EEROM

Location	Description
7E00 }	Stepper Reference Count 1 D1 measurement
7E01 }	
7E02 }	Stepper Reference Count 2 D2 reference
7E03 }	
7E04 }	Stepper Reference Count 3 D3 measurement
7E05 }	
7E06 }	Stepper Reference Count 4 D4 reference
7E07 }	
7E08 }	Stepper Reference Count 5 D5 measurement
7E09 }	
7E0A }	Stepper Reference Count 6 D6 reference
7E0B }	
7E0C }	Stepper Reference Count 7 D7 measurement
7E0D }	
7E0E }	Delay in D1 measurement – range 20 (14h) to 1250 (04E2h) 4usec/unit
7E0F }	
7E10 }	Delay in D1 measurement – range 20 (14h) to 1250 (04E2h) 4usec/unit
7E11 }	
7E12 }	Delay in D1 measurement – range 20 (14h) to 1250 (04E2h) 4usec/unit
7E13 }	
7E14 }	Delay in D1 measurement – range 20 (14h) to 1250 (04E2h) 4usec/unit
7E15 }	
7E16 }	Delay in D1 measurement – range 20 (14h) to 1250 (04E2h) 4usec/unit
7E17 }	
7E18 }	Delay in D1 measurement – range 20 (14h) to 1250 (04E2h) 4usec/unit
7E19 }	
7E1A }	Delay in D1 measurement – range 20 (14h) to 1250 (04E2h) 4usec/unit
7E1B }	
7E1C	D1 gain divisor
7E1D	D1 gain multiplier (msb = 8 for hardware *3 gain)
7E1E	D2 gain divisor
7E1F	D2 gain multiplier (msb = 8 for hardware *3 gain)

e.g.

		7E1C	7E1D		
7E1C/1D	≡	01,	01,	d1	= 5000 (no gain)
		01,	81,	d1	= 15000 (x3 hardware gain only)
		01,	03,	d1	= 15000 (x3 software gain only)
		01,	83,	d1	= 45000 (x3 hardware gain & x3 software gain)

Model: GCEM40x0 IC2

Software ID: 507-056A

Module: MM7E20

Memory Map: Internal EEROM

Location	Description	
7E20	D3 gain divisor	
7E21	D3 gain multiplier (msb = 8 for hardware *3 gain)	
7E22	D4 gain divisor	
7E23	D4 gain multiplier (msb = 8 for hardware *3 gain)	
7E24	D5 gain divisor	
7E25	D5 gain multiplier (msb = 8 for hardware *3 gain)	
7E26	D6 gain divisor	
7E27	D6 gain multiplier (msb = 8 for hardware *3 gain)	
7E28	D7 gain divisor	
7E29	D7 gain multiplier (msb = 8 for hardware *3 gain)	
7E2A	00	}
7E2B	00	}
7E2C	04	}
7E2D	EF	}
7E2E	04	Sampling array size (no. of elements max 2808dec = 0AF8hex)
7E2F	00	([0668h] 16400 at 30.8us = 50.51msec)
7E30	0D	Analysis sample size (no. of elements max 1398dec = 0576hex)
7E31	02	([0514h] 1300 at 30.8us = 40.04msec)
7E32	No. of thermistor digitisations	
7E33	Allowable saturation count	
7E34		
7E35		
7E36		
7E37		
7E38		
7E39		
7E3A		
7E3B		
7E3C		
7E3D		
7E3E		
7E3F	Strobe delay flag (01 = offset strobe by 1 millisec)	

Model: GCEM40x0 IC2

Software ID: 507-056A

Module: MM7E40

Memory Map: Internal EEROM

Location	Description
----------	-------------

7E40	48	}	Detector offset target value
7E41	00	}	
7E42	30		Proportional constant
7E43	20		Integral constant
7E44	01		Control flag (01 for auto, 00 for manual)
7E45	01		Time constant n
7E46	80	}	DAC control value
7E47	00	}	
7E48	01		Detector offset smoothing factor
7E49	FF		
7E4A	03	}	Proportional divider for det. gain (*1)
7E4B	20	}	
7E4C	12	}	Proportional divider for det. gain (*6)
7E4D	10	}	
7E4E	0A	}	Integral divider for det. gain (*1)
7E4F	00	}	
7E50	3C	}	Integral divider for det. gain (*6)
7E51	00	}	

7E52

7E53

7E54

7E55

7E56

7E57

7E58

7E59

7E5A

7E5B

7E5C

7E5D

7E5E

7E5F

Default EEROM >00 = load default EEROM parameters at start-up

DDU MASTER

Model: Gcem40X0 5CH DDU Software ID: D07-107B Module: MM6F00

Memory Map: Internal EEROM :- Misc Data

Location	Description	
6F00	01	00=2ch 'small' 01=5ch 'large' DDU
6F01		
6F02		
6F03		
6F04		
6F05		
6F06		
6F07		
6F08		
6F09		
6F0A		
6F0B		
6F0C		
6F0D		
6F0E		
6F0F		
6F10		
6F11		
6F12		
6F13		
6F14		
6F15		
6F16		
6F17		
6F18		
6F19		
6F1A		
6F1B		
6F1C		
6F1D		
6F1E		
6F1F	00	Set default EPROM data >00 = yes (default eeprom set at power up)

Model: Gcem40X0 5CH DDU Software ID: D07-107A Module: MM6F60

Memory Map: Internal EEROM :- DAC Calibration Data

Location	Description		
6F60	}	DAC span	}
6F61	}		} Ch.1
6F62	}	DAC zero	}
6F63	}		}
6F64	}	DAC span	}
6F65	}		} Ch.2
6F66	}	DAC zero	}
6F67	}		}
6F68	}	DAC span	}
6F69	}		} Ch.3
6F6A	}	DAC zero	}
6F6B	}		}
6F6C	}	DAC span	}
6F6D	}		} Ch.4
6F6E	}	DAC zero	}
6F6F	}		}
6F70	}	DAC span	}
6F71	}		} Ch.5
6F72	}	DAC zero	}
6F73	}		}
6F74	}	DAC span	}
6F75	}		} Ch.6(Not Used)
6F76	}	DAC zero	}
6F77	}		}
6F78	}	DAC span	}
6F79	}		} Ch.7(Not Used)
6F7A	}	DAC zero	}
6F7B	}		}
6F7C	}	DAC span	}
6F7D	}		} Ch.8(Not Used)
6F7E	}	DAC zero	}
6F7F	}		}

Model: Gcem40X0 5CH DDU Software ID: D07-107A Module: MM6F80

Memory Map: Internal EEROM :-

Location	Description	
6F80	Species pointer	} 00=gas ch1,01=gas ch2 etc
6F81	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6F82	0/4mA (00=0mA 01=4mA)	}
6F83	Data invalid	} 00=Zero 01=F.S. 02=Hold 03=Meas
6F84	Averaging value	}
6F85	} Span	} O/P 1
6F86	}	}
6F87	} Zero	}
6F88	}	}
6F89	Species pointer	} 00=gas ch1,01=gas ch2 etc
6F8A	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6F8B	Averaging value	}
6F8C		} Alarm 1
6F8D	} Level	}
6F8E	}	}
6F8F		}
6F90	Species pointer	} 00=gas ch1,01=gas ch2 etc
6F91	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6F92	0/4mA (00=0mA 01=4mA)	}
6F93	Data invalid	} 00=Zero 01=F.S. 02=Hold 03=Meas
6F94	Averaging value	}
6F95	} Span	} O/P 2
6F96	}	}
6F97	} Zero	}
6F98	}	}
6F99	Species pointer	} 00=gas ch1,01=gas ch2 etc
6F9A	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6F9B	Averaging value	}
6F9C		} Alarm 2
6F9D	} Level	}
6F9E	}	}
6F9F		}

Model: Gcem40X0 5CH DDU Software ID: D07-107A Module: MM6FA0

Memory Map: Internal EEROM :-

Location	Description	
6FA0	Species pointer	} 00=gas ch1,01=gas ch2 etc
6FA1	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6FA2	0/4mA (00=0mA 01=4mA)	}
6FA3	Data invalid	} 00=Zero 01=F.S. 02=Hold 03=Meas
6FA4	Averaging value	}
6FA5	} Span	} O/P 3
6FA6	}	}
6FA7	} Zero	}
6FA8	}	}
6FA9	Species pointer	} 00=gas ch1,01=gas ch2 etc
6FAA	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6FAB	Averaging value	}
6FAC		} Alarm 3
6FAD	} Level	}
6FAE	}	}
6FAF		}
6FB0	Species pointer	} 00=gas ch1,01=gas ch2 etc
6FB1	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6FB2	0/4mA (00=0mA 01=4mA)	}
6FB3	Data invalid	} 00=Zero 01=F.S. 02=Hold 03=Meas
6FB4	Averaging value	}
6FB5	} Span	} O/P 4
6FB6	}	}
6FB7	} Zero	}
6FB8	}	}
6FB9	Species pointer	} 00=gas ch1,01=gas ch2 etc
6FBA	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6FBB	Averaging value	}
6FBC		} Alarm 4
6FBD	} Level	}
6FBE	}	}
6FBF		}

Model: Gcem40X0 5CH DDU Software ID: D07-107A Module: MM6FC0

Memory Map: Internal EEROM :-

Location	Description	
6FC0	Species pointer	} 00=gas ch1,01=gas ch2 etc
6FC1	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6FC2	0/4mA (00=0mA 01=4mA)	}
6FC3	Data invalid	} 00=Zero 01=F.S. 02=Hold 03=Meas
6FC4	Averaging value	}
6FC5	} Span	} O/P 5
6FC6	}	}
6FC7	} Zero	}
6FC8	}	}
6FC9	Species pointer	} 00=gas ch1,01=gas ch2 etc
6FCA	Units	} 00=vpm 01=mg/m3 02=mg/Nm3
6FCB	Averaging value	}
6FCC		} Alarm 5
6FCD	} Level	}
6FCE	}	}
6FCF		}
6FD0	} Absolute data loc.	}
6FD1	}	}
6FD2	0/4mA (00=0mA 01=4mA)	}
6FD3	Data invalid	} 00=Zero 01=F.S. 02=Hold 03=Meas
6FD4	Averaging value	}
6FD5	} Span	} O/P 6(Not Used)
6FD6	}	}
6FD7	} Zero	}
6FD8	}	}
6FD9	} Absolute data loc.	}
6FDA	}	}
6FDB	Averaging value	}
6FDC		} Alarm 6(Not Used)
6FDD	} Level	}
6FDE	}	}
6FDF	Source (00=1byte)	}

Model: Gcem40X0 5CH DDU Software ID: D07-107A Module: MM6FE0

Memory Map: Internal EEROM :-

Location	Description
6FE0	} Absolute data loc. }
6FE1	} }
6FE2	} 0/4mA (00=0mA 01=4mA) }
6FE3	} Data invalid }
6FE4	} Averaging value }
6FE5	} Span }
6FE6	} }
6FE7	} Zero }
6FE8	} }
6FE9	} Absolute data loc. }
6FEA	} }
6FEB	} Averaging value }
6FEC	} Alarm 7(Not Used) }
6FED	} Level }
6FEE	} }
6FEF	} Source (00=1byte) }
6FF0	} Absolute data loc. }
6FF1	} }
6FF2	} 0/4mA (00=0mA 01=4mA) }
6FF3	} Data invalid }
6FF4	} Averaging value }
6FF5	} Span }
6FF6	} }
6FF7	} Zero }
6FF8	} }
6FF9	} Absolute data loc. }
6FFA	} }
6FFB	} Averaging value }
6FFC	} Alarm 8(Not Used) }
6FFD	} Level }
6FFE	} }
6FFF	} Source (00=1byte) }

Model: Gcem40X0 5ch DDU Software ID: 507-107A Module: MM0820

Sub-Routine: Parameter Data

Location	Description
0820 }	Temp. 7ED8 }
0821 }	Plant status thresholds }
0822 }	Plant status source }
0823 }	Pathlength mm 7F30 }
0824 }	
0825 }	
0826 }	NO2 % [32h =5.0%]
0827 }	
0828 }	Temperature }
0829 }	Oxygen }
082A }	Pressure }
082B }	Water vapour }
082C }	Temperature }
082D }	Oxygen }
082E }	Pressure }
082F }	Water vapour }
0830 }	Temperature }
0831 }	
0832 }	Oxygen }
0833 }	Normalisation Keypad Inputs }
0834 }	Pressure }
0835 }	
0836 }	Water Vapour }
0837 }	
0838	
0839	
083A	
083B	
083C	
083D	
083E	
083F	

Model: Gcem40X0 DDU Software ID: D07-107B Module: MM0840

Sub-Routine: Parameter data

Location	Description	
0840	Pointer for 'wet' or 'dry' O2 measurement (not used) 00=dry, 01=wet	
0841	Cal interval (hours)	
0842	1st cal time (hours)	
0843	Cal initiate:- logic (if clear use plant status)	
0844	Cal alarm:- if clear then alarm = cal in progress	
0845	Cal data to output:- if clear then YES	
0846	Zero purge 00=OFF, 01=ON	
0847	Span gas 00=OFF, 01=ON	[not yet fully implemented]
0848	Auto Span gas sequence 00=OFF, 01=ON	[not yet implemented]
0849		
084A	After Cal Delay mins	
084B	Purge time mins	
084C }	Target cal value	
084D }		
084E	Number of target cal cycles	
084F	Cal Initiate Flag	
0850 }		
0851 }	Scroll parameters	
0852 }		
0853 }		
0854 }	Security Code	
0855 }		
0856 }	1st time start up code 5555	
0857 }		
0858	O2 or CO2 normalisation 00=Yes, 01=inhibited	[not yet implemented]
0859	NO or NOX display 00=NO, 01=NOX	
085A	O2 or CO2 normalisation 00=O2, 01=CO2	
085B	O2 source 00=DDU 01=AWC	
085C	Maintenance flag	
085D	Codel/ModBus serial comms address	
085E	Protocol flag 00=Codel 01=ModBus	
085F	Baud Rate flag 00=4800 01=9600	

Model: Gcem40X0 5CH DDU Software ID: D07-107A Module: MM0900

Memory Map : Output Data Block (Rolling ave's can be zero'ed if plant off)

Location	Description		
0900 }	vpm or %vol	}	
0901 }		}	(4080)
0902 }	mg/m3	}	Channel 1 (NO)
0903 }		}	
0904 }	mg/Nm3	}	
0905 }		}	
0906 }	vpm or %vo	}	
0907 }		}	
0908 }	mg/m3	}	Channel 2 (SO2)
0909 }		}	
090A }	mg/Nm3	}	
090B }		}	
090C }	vpm or %vol	}	
090D }		}	
090E }	mg/m3	}	Channel 3 (CO)
090F }		}	
0910 }	mg/Nm3	}	
0911 }		}	
0912 }	vpm or %vol	}	
0913 }		}	
0914 }	mg/m3	}	Channel 4 (CO2)
0915 }		}	
0916 }	mg/Nm3	}	
0917 }		}	
0918 }	vpm or %vol	}	
0919 }		}	
091A }	mg/m3	}	Channel 5 (H2O)
091B }		}	
091C }	mg/Nm3	}	
091D }		}	
091E }	Temperature	}	
091F }		}	Normalisation Data

Model: Gcem40X0 DDU Software ID: 507-106A Module: MM7900

Memory Map: External RAM :- Copy of dignostic data for IEM 7900-79FF

Location	Description
7900 }	
7901 }	D1
7902 }	
7903 }	
7904 }	
7905 }	D2
7906 }	
7907 }	
7908 }	
7909 }	D3
790A }	
790B }	
790C }	
790D }	D4
790E }	
790F }	
7910 }	
7911 }	D5
7912 }	
7913 }	
7914]	Anticipated changes
7915]	D6 reserved
7916]	
7917]	
7918]	
7919]	D7 reserved
791A]	
791B]	
791C }	PbSe Saturation counts
791D }	
791E }	
791F }	

Data Register Locations at CODEL Gcem40x0 DDU for interrogation using Standard MODBUS

Subject : Preliminary

Doc code : Blank

Issue : A

Date : 04/06/2009

CODEL Doc ref: QA - 72 B

REGISTER FORMAT					CUSTOMER SPECIFIC DETAILS			Description
Location		Analyser Data	Data	Units	Typical Range	Tag	Value	
0000	D1 High word	instantaneous	XXXX					
0001	D1 Low word	instantaneous	XXXX	Long word	0-8388607		XXXX	
0002	D2 High word	instantaneous	XXXX				XXXX	
0003	D2 Low word	instantaneous	XXXX	Long word	0-8388607		XXXX	
0004	D3 High word	instantaneous	XXXX				XXXX	
0005	D3 Low word	instantaneous	XXXX	Long word	0-8388607		XXXX	
0006	D4 High word	instantaneous	XXXX				XXXX	
0007	D4 Low word	instantaneous	XXXX	Long word	0-8388607		XXXX	
0008	D5 High word	instantaneous	XXXX				XXXX	
0009	D5 Low word	instantaneous	XXXX	Long word	0-8388607		XXXX	
000A	D6 High word (reserved)	instantaneous	XXXX				XXXX	
000B	D6 Low word (reserved)	instantaneous	XXXX	Long word			XXXX	
000C	D7 High word (reserved)	instantaneous	XXXX				XXXX	
000D	D7 Low word (reserved)	instantaneous	XXXX	Long word			XXXX	
000E	PbSe saturation counts	instantaneous	XXXX	logic	0000=valid, else=invalid		XXXX	
000F	not used		XXXX				XXXX	
0010	Thermistor	instantaneous	XXXX	integers	0-32767		XXXX	
0011	Cooler Current	instantaneous	XXXX	0.1 mA	0-1500.0		XXXX	
0012	Source Heatsink Temperature	instantaneous	XXXX	0.01 degree C +40	0-190.00		XXXX	
0013	Motor PCB Temperature	instantaneous	XXXX	0.01 degree C +40	0-190.00		XXXX	
0014	Micro PCB Temperature	instantaneous	XXXX	0.01 degree C +40	0-190.00		XXXX	
0015	Heater Plate Temperature	instantaneous	XXXX	0.01 degree C +40	0-190.00		XXXX	
0016	Lens Holder Temperature	instantaneous	XXXX	0.01 degree C +40	0-190.00		XXXX	
0017	Detector Heatsink Temperature	instantaneous	XXXX	0.01 degree C +40	0-190.00		XXXX	
0018	Source voltage	instantaneous	XXXX	0.001 volts	0-5.000		XXXX	
0019	Cooler Drive voltage	instantaneous	XXXX	0.001 volts	0-5.000		XXXX	
001A	not used		XXXX				XXXX	
001B	not used		XXXX				XXXX	

001C	not used		XXXX			XXXX
001D	not used		XXXX			XXXX
001E	not used		XXXX			XXXX
001F	Gcem monitor valid flags	instantaneous	XXXX	logic	XX00=data valid, else=invalid	XXXX
0020	D1 High word	smoothed	XXXX			XXXX
0021	D1 Low word	smoothed	XXXX	Long word	0-8388607	XXXX
0022	D2 High word	smoothed	XXXX			XXXX
0023	D2 Low word	smoothed	XXXX	Long word	0-8388607	XXXX
0024	D3 High word	smoothed	XXXX			XXXX
0025	D3 Low word	smoothed	XXXX	Long word	0-8388607	XXXX
0026	D4 High word	smoothed	XXXX			XXXX
0027	D4 Low word	smoothed	XXXX	Long word	0-8388607	XXXX
0028	D5 High word	smoothed	XXXX			XXXX
0029	D5 Low word	smoothed	XXXX	Long word	0-8388607	XXXX
002A	D6 Low word (reserved)	smoothed	XXXX			XXXX
002B	D6 Low word (reserved)	smoothed	XXXX	Long word		XXXX
002C	D7 Low word (reserved)	smoothed	XXXX			XXXX
002D	D7 Low word (reserved)	smoothed	XXXX	Long word		XXXX
002E	PbSe saturation counts	instantaneous	XXXX	logic	0000=valid, else=invalid	XXXX
002F	PT100 temperature	smoothed	XXXX	0.01 degree C	0-200.00	XXXX
0030	Thermistor	smoothed	XXXX	integers	0-32767	XXXX
0031	Cooler Current	smoothed	XXXX	0.1 mA	0-1500.0	XXXX
0032	Source Heatsink Temperature	smoothed	XXXX	0.01 degree C +40	0-190.00	XXXX
0033	Motor PCB Temperature	smoothed	XXXX	0.01 degree C +40	0-190.00	XXXX
0034	Micro PCB Temperature	smoothed	XXXX	0.01 degree C +40	0-190.00	XXXX
0035	Heater Plate Temperature	smoothed	XXXX	0.01 degree C +40	0-190.00	XXXX
0036	Lens Holder Temperature	smoothed	XXXX	0.01 degree C +40	0-190.00	XXXX
0037	Detector Heatsink Temperature	smoothed	XXXX	0.01 degree C +40	0-190.00	XXXX
0038	Source voltage	smoothed	XXXX	0.001 volts	0-5.000	XXXX
0039	Cooler Drive voltage	smoothed	XXXX	0.001 volts	0-5.000	XXXX
003A	Heater Plate Temperature M:S drive ratio	instantaneous	XXXX	integers	0-65535	XXXX
003B	Detector Heatsink Temperature M:S drive ratio	instantaneous	XXXX	integers	0-65535	XXXX
003C	not available		XXXX			XXXX
003D	not available		XXXX			XXXX
003E	not available		XXXX			XXXX
003F	not available		XXXX			XXXX

0040	not available		XXXX			XXXX
0041	not available		XXXX			XXXX
0042	not available		XXXX			XXXX
0043	not available		XXXX			XXXX
0044	not available		XXXX			XXXX
0045	not used		XXXX			XXXX
0046	not used		XXXX			XXXX
0047	Channel 1 Y value	instantaneous	XXXX	integers	0-32767	XXXX
0048	Channel 2 Y value	instantaneous	XXXX	integers	0-32768	XXXX
0049	Channel 3 Y value	instantaneous	XXXX	integers	0-32769	XXXX
004A	Channel 4 Y value	instantaneous	XXXX	integers	0-32770	XXXX
004B	Channel 5 Y value (reserved)	instantaneous	XXXX	integers	0-32771	XXXX
004C	Channel 6 Y value (reserved)	instantaneous	XXXX	integers	0-32772	XXXX
004D	not used		XXXX			XXXX
004E	Channel 1 Z value	instantaneous	XXXX	integers	0-32774	XXXX
004F	Channel 2 Z value	instantaneous	XXXX	integers	0-32775	XXXX
0050	Channel 3 Z value	instantaneous	XXXX	integers	0-32776	XXXX
0051	Channel 4 Z value	instantaneous	XXXX	integers	0-32777	XXXX
0052	Channel 5 Z value (reserved)	instantaneous	XXXX	integers	0-32778	XXXX
0053	Channel 6 Z value (reserved)	instantaneous	XXXX	integers	0-32779	XXXX
0054	not used		XXXX			XXXX
0055	Channel 1 Z value	smoothed	XXXX	integers	0-32781	XXXX
0056	Channel 2 Z value	smoothed	XXXX	integers	0-32782	XXXX
0057	Channel 3 Z value	smoothed	XXXX	integers	0-32783	XXXX
0058	Channel 4 Z value	smoothed	XXXX	integers	0-32784	XXXX
0059	Channel 5 Z value (reserved)	smoothed	XXXX	integers	0-32785	XXXX
005A	Channel 6 Z value (reserved)	smoothed	XXXX	integers	0-32786	XXXX
005B	not used		XXXX			XXXX
005C	not used		XXXX			XXXX
005D	not used		XXXX			XXXX
005E	not used		XXXX			XXXX
005F	not used		XXXX			XXXX
0060	Active Weather Cover Temperature	smoothed	XXXX	0.01 degree C +40	0-190.00	XXXX
0061	Active Weather Cover Temperature Target Value	constant	XXXX	0.01 degree C +40	0-190.00	XXXX
0062	not available		XXXX			XXXX

0063	not available		XXXX				XXXX	G-CEM 40x0 NO G-CEM 40x0 SO2 G-CEM 40x0 H2O G-CEM 40x0 CO2 G-CEM 40x0 CO
0064	Active Weather Cover Temperature M:S drive ratio		XXXX	integers	0-65535		XXXX	
0065	not available		XXXX				XXXX	
0066	not available		XXXX				XXXX	
0067	not available		XXXX				XXXX	
0068	not used		XXXX				XXXX	
0069	not used		XXXX				XXXX	
006A	not used		XXXX				XXXX	
006B	not used		XXXX				XXXX	
006C	not used		XXXX				XXXX	
006D	not used		XXXX				XXXX	
006E	not used		XXXX				XXXX	
006F	not used		XXXX				XXXX	
0070	G-CEM 40x0 channel 1 NO	1	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0071	G-CEM 40x0 channel 2 SO2	1	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0072	G-CEM 40x0 channel 3 H2O	1	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0073	G-CEM 40x0 channel 4 CO2	1	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0074	G-CEM 40x0 channel 5 CO	1	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0075	not used	1	XXXX				XXXX	
0076	not used	1	XXXX				XXXX	
0077	Normalisation Source - Temperature	1	XXXX	1 degree C	0-300		XXXX	
0078	Normalisation Source Data - Oxygen	1	XXXX	0.1%	0-25.0		XXXX	
0079	Normalisation Source Data - Pressure	1	XXXX	1KPa	0-160		XXXX	
007A	Normalisation Source Data - Water Vapour	1	XXXX	0.1%	0-50.0		XXXX	
007B	Flue Gas Temperature	1	XXXX	1 degree C	0-300		XXXX	
007C	Data Valid	1	XXXX	logic	XX00=valid, else=invalid		XXXX	
007D	Purging/Calibration Status	1	XXXX	logic	00XX=purge off, XX00=Cal off		XXXX	
007E	Plant Status/Head Identifier	1	XXXX	logic	000A=plant on, else Plant Off		XXXX	
007F	G-CEM 40x0 DDU Comms validation	1	XXXX	logic	00XX=valid, FFX=no reply		XXXX	
0400	G-CEM 4000 channel 1 - x min	1	XXXX	1vpm (or 0.1%)	0-3000vpm (0-50.0% if CO2 or H2O)		XXXX	
0401	G-CEM 4000 channel 1 - x min	1	XXXX	mg/m3	0-9999mg/m3(0-50.0% if CO2 or H2O)		XXXX	
0402	G-CEM 4000 channel 1 - x min	1	XXXX	mg/Nm3	0-9999mg/Nm3(0-50.0% if CO2 or H2O)		XXXX	
0403	G-CEM 4000 channel 2 - x min	1	XXXX	1vpm (or 0.1%)	0-3000vpm (0-50.0% if CO2 or H2O)		XXXX	
0404	G-CEM 4000 channel 2 - x min	1	XXXX	mg/m3	0-9999mg/m3(0-50.0% if CO2 or H2O)		XXXX	

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0405	G-CEM 4000 channel 2 - x min	1	XXXX	mg/Nm3	0-9999mg/Nm3(0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x5 SO2
0406	G-CEM 4000 channel 3 - x min	1	XXXX	0.1% vol.	0-3000vpm (0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x6 H2O
0407	G-CEM 4000 channel 3 - x min	2	XXXX	0.1% vol.	0-9999mg/Nm3(0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x7 H2O
0408	G-CEM 4000 channel 3 - x min	1	XXXX	0.1% vol.	0-9999mg/Nm3(0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x8 H2O
0409	G-CEM 4000 channel 4 - x min	1	XXXX	0.1% vol.	0-3000vpm (0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x9 CO2
040A	G-CEM 4000 channel 4 - x min	1	XXXX	0.1% vol.	0-9999mg/Nm3(0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x10 CO2
040B	G-CEM 4000 channel 4 - x min	1	XXXX	0.1% vol.	0-9999mg/Nm3(0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x11 CO2
040C	G-CEM 4000 channel 5 - x min	1	XXXX	0.1% vol.	0-3000vpm (0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x12 CO
040D	G-CEM 4000 channel 5 - x min	1	XXXX	0.1% vol.	0-9999mg/Nm3(0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x13 CO
040E	G-CEM 4000 channel 5 - x min	1	XXXX	0.1% vol.	0-9999mg/Nm3(0-50.0% if CO2 or H2O)	XXXX	G-CEM 40x14 CO
040F	Normalisation Data - Temperature	1	XXXX	1 degree C	0-300	XXXX	G-CEM 40x15 norm temp
0410	Normalisation Data - Oxygen/CO2 dry	1	XXXX	0.1%	0-25.0	XXXX	G-CEM 40x16 norm O2/CO2
0411	Normalisation Data - Pressure	1	XXXX	1KPa	0-160	XXXX	G-CEM 40x17 norm press
0412	Normalisation Data - Water Vapour	1	XXXX	0.1%	0-50.0	XXXX	G-CEM 40x18 norm H2O